Method overriding

Method Overriding in Java

- **Definition:**
  Method overriding is an Object-Oriented Programming (OOP) concept in which a **subclass provides its own implementation of a method that is already defined in its superclass**. The overridden method must have the **same method name, same parameters, and same return type** as the method in the parent class.
- It is mainly used to achieve **runtime polymorphism (dynamic method dispatch)**.

Rules for Method Overriding

- The method name must be the same.
- The parameter list must be identical.
- The return type should be the same (or covariant).
- The method in the subclass cannot have more restrictive access than the parent method.
- Static, final, and private methods cannot be overridden.

Advantages

- Supports runtime polymorphism.
- Improves code flexibility and reusability.
- Allows different classes to implement behavior according to their needs.
- Makes programs easier to maintain and extend.

```
class Animal
 {
    void sound()
          {
        System.out.println("Animals make different sounds.");
          }
}

class Dog extends Animal
{
          void sound()
          {
        System.out.println("Dog barks: Bow Bow");
          }
}

class Cat extends Animal
{
          void sound()
          {
        System.out.println("Cat meows: Meow Meow");
           }
}
```

```
public class Demo
{
    public static void main(String[] args)
        {

        Animal  a = new Dog();
         a.sound();

       Animal  c= new Cat();
         c.sound();
         }
}
```

Output

Dog barks: Bow Bow

Cat meows: Meow Meow